

Assignment 1 Submission; due Thursday, September 24, 2026

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- [1] Let $E \subseteq \mathbb{R}$ be nonempty. We say that $t \in \mathbb{R}$ is a *lower bound* of E if $t \leq x$ for all $x \in E$. We say that E is *bounded below* if there exists a lower bound of E . We say that $t \in \mathbb{R}$ is a *greatest lower bound*, or *infimum*, of E if it is a lower bound of E and if $t \geq s$ for any lower bound s of E . If E has an infimum, it is denoted $\inf E$.

- [a] Let E be a nonempty subset of $\mathbb{R}$ which is bounded below. Find a way to use the completeness axiom to prove that $\inf E$ exists. *Hint:* Consider the subset $-E = \{-x : x \in E\}$.

Answer:

Proof. Since E is nonempty and bounded below, there exists $\alpha \in \mathbb{R}$ such that

$$\forall x \in E : \alpha \leq x$$

Consequently,

$$\forall x \in E : -\alpha \geq -x$$

.

Define

$$-E = \{-x : x \in E\}.$$

Then, $-\alpha$ is an upper bound of $-E$.

By the completeness axiom, $\sup(-E)$ exists. Let

$$\beta = \sup(-E).$$

It follows that $\forall x \in -E : \beta \geq x$. And therefore, $\forall x \in E : -\beta \leq x$. So $-\beta$ is a lower bound of E .

Next, take any other lower bound of E , call it s . From the same logic as above, it follows that $-s$ is an upper bound of $-E$. Since β is a supremum of $-E$, it follows that $\beta \leq -s$.

So, $-\beta \geq s$. Therefore, since $-\beta$ is a lower bound of E and it is at least as large as any other lower bound, it follows that $-\beta = \inf E$ $\square$

- [b] Let $\emptyset \neq A \subseteq B \subseteq \mathbb{R}$. Suppose B is bounded below (so $\inf B$ exists). Prove that A is bounded below (so $\inf A$ exists) and that $\inf A \geq \inf B$.

Answer:

Proof. If B is bounded below, then $\exists \alpha : \forall x \in B : \alpha \leq x$.

Since $A \subseteq B$, we know that $\forall y \in A : y \in B$.

Consequently,

$$\forall y \in A : \alpha \leq y$$

Therefore, α is also a lower bound of A . And from 1[a], we can conclude that $\inf A$ also exists.

Now, since A is a subset of B and $\inf B$ is a lower bound of B , then $\inf B$ must also be a lower bound of A . Since $\inf A$ must be greater than or equal to any other lower bounds of A , we have that

$$\inf A \geq \inf B$$

□

- [c] Let A, B be nonempty subsets of $\mathbb{R}$, both bounded below (so both $\inf A$ and $\inf B$ exist). Define

$$A + B = \{a + b : a \in A, b \in B\}.$$

Prove that $A + B$ is bounded below and that $\inf(A + B) = \inf A + \inf B$.

Answer:

Proof.

$$\begin{aligned} \forall a \in A : \inf A &\leq a \\ \forall b \in B : \inf B &\leq b \\ \therefore \forall a \in A, b \in B : \inf A + \inf B &\leq a + b \\ \therefore \forall x \in A + B : \inf A + \inf B &\leq x \end{aligned}$$

So $\inf A + \inf B$ is a lower bound of $A + B$. Consequently, $\inf(A + B)$ exists.

Now, we know that

$$\begin{aligned} \forall \varepsilon > 0 : \exists \alpha \in A : \inf A + \frac{\varepsilon}{2} &> \alpha \\ \forall \varepsilon > 0 : \exists \beta \in B : \inf B + \frac{\varepsilon}{2} &> \beta \end{aligned}$$

Therefore,

$$\forall \varepsilon > 0 : \exists \alpha \in A, \beta \in B : \inf A + \inf B + \varepsilon > \alpha + \beta$$

This means that no number greater than $\inf A + \inf B$ can be a lower bound of $A + B$. Therefore,

$$\inf(A + B) = \inf A + \inf B$$

□

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- [2] Let x, y be two real numbers such that $y - x > 1$. (In particular, $x < y$.)

- [a] If x is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. By the Archimedean property, the set $E = \{n \in \mathbb{N} : n > x\} \subset \mathbb{N} \subset \mathbb{Z}$ is not empty.

x is a lower bound of E , so by the completeness axiom, $\inf E$ exists. And since $E \subset \mathbb{Z}$, $\inf E \in E$. Let $\inf E = k$.

Then $x < k$. Now, assume for the sake of contradiction that $x < k - 1$. Since $k - 1 \in \mathbb{Z}$, then $k - 1 \in E$. But $k - 1 < k$ and $k = \inf E$, which is a contradiction.

Therefore, $k - 1 \leq x$. $k \leq x + 1 < y$.

So, $x < k < y$.

□

- [b] If x is negative and y is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. Take $k = 0 \in \mathbb{Z}$.

□

- [c] If y is negative, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. Since $y < 0$, we have $-y > 0$. Moreover,

$$(-x) - (-y) = y - x > 1$$

So applying 2[a] to $-x$ & $-y$, we get that $\exists n \in \mathbb{Z} : -y < n < -x$.

$$x < -n < y$$

Taking $k = -n \in \mathbb{Z}$ proves the result.

□

- [3] The parts of this problem are *not related*.

- [a] Let $x \in \mathbb{R}$. Consider the set $E = \{q \in \mathbb{Q} : q < x\} \subset \mathbb{Q} \subset \mathbb{R}$. Show that E is nonempty and bounded above, so $\sup E$ exists. Show also that $\sup E = x$.

Answer:

- [b] Let $x, y \in \mathbb{R}$ with $x < y$. Prove that there exists an *irrational* (not rational) number z such that $x < z < y$. *Hint:* You may use the fact that $\sqrt{2}$ is irrational.

Answer:

[4] Let $C > 0$ be a fixed positive number.

[a] Let $x \geq 0$, and suppose that for any $\varepsilon > 0$, we have $x < C\varepsilon$. Show that for any $\eta > 0$, we have $x < \eta$.

Answer:

[b] In the situation of part [a], show that we must have $x = 0$. *Hint:* Find a way to use the Archimedean property.

Answer:

[5] COMING SOON

Answer:

[6] COMING SOON

Answer:

[7] COMING SOON

Answer:

[8] COMING SOON

Answer:

[9] COMING SOON

Answer:

[10] COMING SOON

Answer: